Generative AI and Large Language Models

Generative artificial intelligence refers to systems capable of producing new content, including text, images, audio, and video, that resembles data they were trained on. Large language models, a prominent category of generative AI, are trained on vast collections of text drawn from books, articles, and websites, learning statistical patterns of language that allow them to generate coherent and contextually relevant text. These models are typically built using transformer architectures and trained through a process where the model learns to predict the next word in a sequence given the preceding context. Once pretrained, large language models can be fine-tuned or prompted to perform a wide variety of tasks, including answering questions, summarizing documents, writing code, and engaging in conversation. Beyond text, generative AI has also made significant progress in image generation, with diffusion models capable of producing highly realistic and creative images from text descriptions. These capabilities have found applications across industries, from content creation and marketing to software development and scientific research assistance. However, generative AI also raises important concerns, including the potential to produce misleading or fabricated information, sometimes referred to as hallucination, as well as questions around copyright when models are trained on existing creative works. The environmental cost of training extremely large models, which require substantial computational resources and energy, has also become a topic of discussion. As generative AI continues to advance, ongoing research focuses on improving factual accuracy, reducing harmful outputs, and developing appropriate safeguards for responsible use.